Computer Networks
Fall 2024/25
SOLUTION
Exercise-4-solution

Problem 1 (20 points).

Fragments (packet) number	Identification number	Flag	Offset	Size of the fragment (Including IP headers)
1	**1900**	**1**	**0**	**700**
2	**1900**	**1**	**680**	**700**
3	**1900**	**1**	**1360**	**700**
4	**1900**	**1**	**2040**	**700**
5	**1900**	**1**	**2720**	**700**
6	**1900**	**0**	**3400**	**80**

Problem 2 (20 points).

Question #	Destination IP address	Answer (one letter)
1	32.2.3.4	J
2	196.94.2.116	B
3	196.94.2.129	C
4	139.1.1.1	H
5	196.94.3.115	A
6	64.111.2.0	I

Problem 3 (20 points).
2,684,354 bps.

Problem 4 (20 points).
1.
A: 2 71 B: 2 4 C: 70 3
3 70 5 1 72 1
2.
A: 2 71 B: 2 inf C: 70 3
3 70 inf 1 72 1
3. around 66 rounds (without poisoned reverse). **[between 60 to 70 is OK.]**
4. 3 rounds (with poisoned reverse). **[between 2 to 4 is OK.]**

Problem 5 (20 points).

Organization	First address	Last address	How many	Mask
A	108.24.208.0	108.24.209.255	**512**	108.24.208.0/23
B	108.24.224.0	108.24.239.255	**4096**	108.24.224.0/20
C	108.25.0.0	108.25.63.255	**16384**	108.25.0.0/18
D	108.24.240.0	108.24.255.255	**4096**	108.24.240.0/20
E	108.24.216.0	108.24.223.255	**2048**	108.24.216.0/21
F	108.24.212.0	108.24.215.255	**1024**	108.24.212.0/22
G	108.25.64.0	108.25.67.555	**1024**	108.25.64.0/22
H	108.24.210.0	108.24.211.255	**512**	108.24.210.0/23